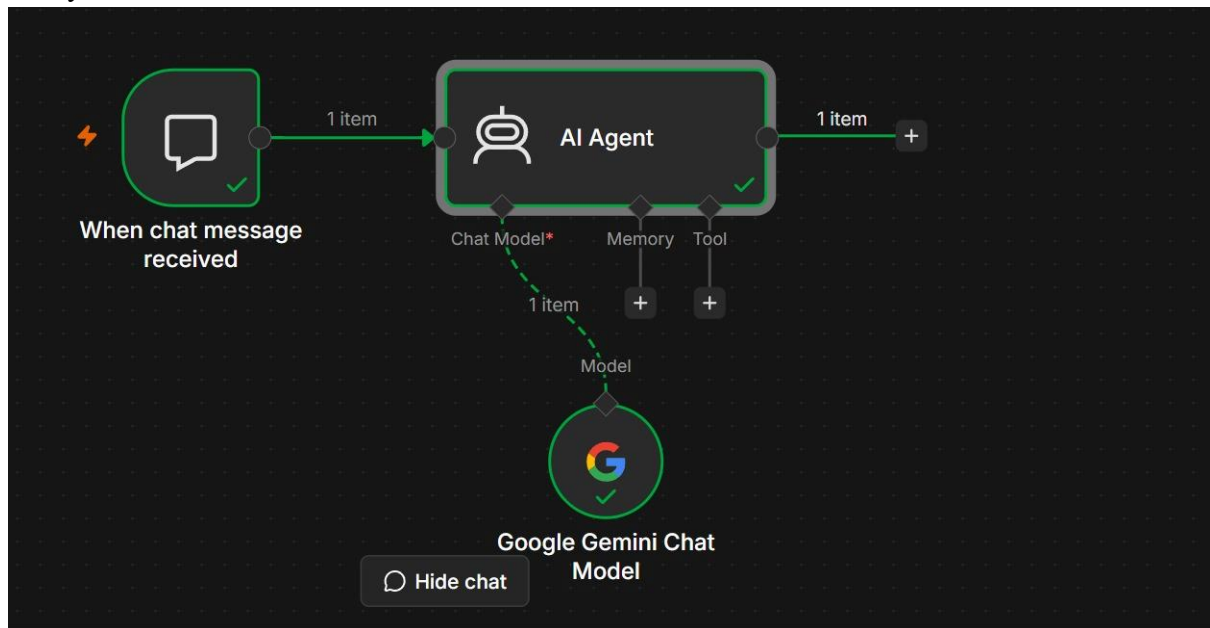


PROJECT-30

AI Python Tutor



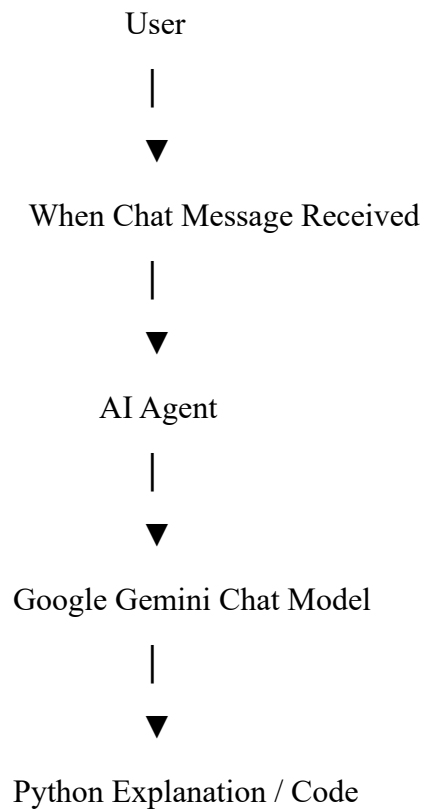
Project Objective

The objective of this project is to build an **AI-powered Python Tutor** that helps users learn Python programming by answering questions, explaining concepts, generating code, debugging programs, and providing step-by-step solutions. The AI Agent acts as a personal Python instructor, making programming easier for beginners and professionals.

The AI Agent can help with:

- Python basics
- Variables and Data Types
- Conditional Statements
- Loops
- Functions
- Object-Oriented Programming (OOP)
- File Handling
- Exception Handling
- Python Libraries
- Debugging Python Code

Workflow Architecture



Workflow Components

Step	Node	Purpose
1	When Chat Message Received	Receives the user's Python programming question
2	AI Agent	Understands the learning request and prepares the prompt
3	Google Gemini Chat Model	Generates Python explanations and code examples

Prerequisites

Before building this project, ensure you have:

- n8n (Cloud or Self-hosted)
 - Google AI Studio API Key
 - Google Gemini Chat Model configured
 - Internet connection
-

Step 1 – Create Workflow

Open **n8n**

Click

New Workflow

Rename it

AI Python Tutor

Step 2 – Add Chat Trigger

Add

When Chat Message Received

Purpose

Receives the user's Python learning request.

Example Input

Explain Python functions with an example.

Output

```
{  
  "chatInput": "Explain Python functions with an example."  
}
```

Step 3 – Add AI Agent

Connect

When Chat Message Received

|



AI Agent

Configure

Source for Prompt

Connected Chat Trigger Node

Prompt

```
{{ $json.chatInput }}
```

Step 4 – Configure the System Prompt

Paste the following into the **System Message**.

You are an AI Python Tutor.

Your task is to teach Python programming in a clear, simple, and interactive manner.

Instructions:

- Carefully understand the user's Python-related question.
 - Explain Python concepts in beginner-friendly language.
 - Generate clean, well-formatted Python code examples.
 - Explain the code line by line whenever appropriate.
 - Follow Python best practices and coding standards.
 - If the user provides code, identify errors and explain how to fix them.
 - Suggest improvements and alternative approaches when applicable.
 - Encourage learning by providing tips and best practices.
 - Return only the explanation and Python code examples.
-

Step 5 – Add Google Gemini Chat Model

Click

+ **Chat Model**

Choose

Google Gemini Chat Model

Configuration

Setting	Value
Provider	Google Gemini

Setting	Value
Model	Gemini 2.5 Flash (or another available Gemini model)
Authentication	Google AI Studio API Key
Connect it to the AI Agent .	

Final Workflow

When Chat Message Received

|



AI Agent

|



Google Gemini Chat Model

Workflow Execution

Step 1

User enters

Explain Python dictionaries with examples.



Step 2

The **When Chat Message Received** node receives the user's request.



Step 3

The **AI Agent** analyzes the learning request and prepares the prompt.



Step 4

The prompt is sent to the **Google Gemini Chat Model**.



Step 5

Gemini generates a detailed explanation along with Python code examples.



Step 6

The explanation and sample code are returned to the user.

Sample Inputs and Outputs

Example 1

Input

Explain Python variables.

Output

Explanation

Variables are used to store data in Python. You can assign values using the assignment operator (=).

```
name = "Alice"
```

```
age = 25
```

```
print(name)
```

```
print(age)
```

Example 2

Input

Write a Python function to calculate the factorial of a number.

Output

```
def factorial(n):
```

```
    if n == 0 or n == 1:
```

```
        return 1
```

```
    return n * factorial(n - 1)
```

```
print(factorial(5))
```

Explanation

This function uses recursion to calculate the factorial of a given number.

Example 3

Input

Explain Python lists.

Output

Explanation

A list is an ordered and mutable collection in Python.

```
fruits = ["Apple", "Banana", "Mango"]
```

```
print(fruits)
```

```
print(fruits[0])
```

```
fruits.append("Orange")
```

```
print(fruits)
```

Example 4

Input

Debug the following Python code:

```
for i in range(5)
```

```
    print(i)
```

Output

Issue

The for statement is missing a colon (:).

Correct Code

```
for i in range(5):
```

```
    print(i)
```

Workflow Diagram

User

|



When Chat Message Received

|



AI Agent

|



Google Gemini Chat Model

|



Python Explanation / Code

Error Handling

Error: No Prompt Specified

Cause

The AI Agent did not receive the user's request.

Solution

Verify that the **Prompt** field contains:

```
{{ $json.chatInput }}
```

Error: Authentication Failed

Cause

Invalid Google Gemini API Key.

Solution

Reconnect or update your Google AI Studio credentials.

Error: Too Many Requests

Cause

Gemini API rate limit exceeded.

Solution

- Wait a few minutes.
- Check your API quota.
- Retry the request.

Learning Outcomes

After completing this project, you will be able to:

- Build an AI-powered Python Tutor in n8n.
- Configure an AI Agent with Google Gemini.
- Design prompts for Python learning and code generation.
- Explain and debug Python programs using AI.
- Build conversational AI workflows for programming education.

Real-World Applications

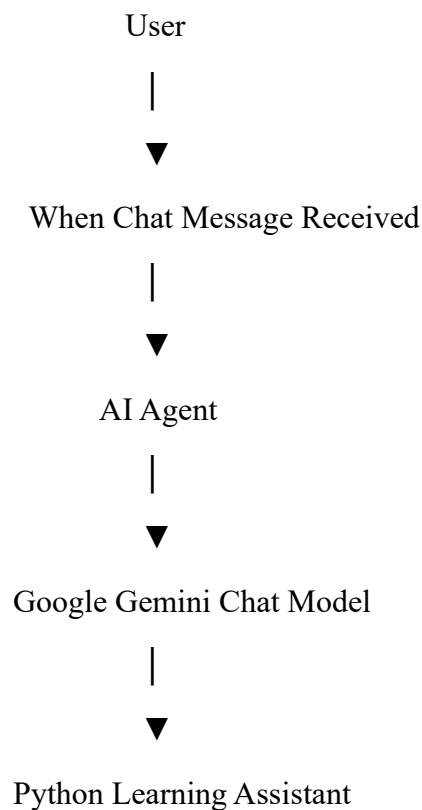
- Python learning assistant
 - Coding education platforms
 - Programming bootcamps
 - Student learning support
 - Code debugging assistant
 - Technical interview preparation
 - Developer productivity tools
 - Online coding tutors
 - AI-powered education platforms
 - Software development training
-

Practice Challenge

Enhance this project by adding:

1. A Python topic selector (Basics, OOP, File Handling, Libraries, etc.).
2. A difficulty level selector (Beginner, Intermediate, Advanced).
3. A code debugging mode for uploaded Python programs.
4. A Memory node to remember the user's learning progress.
5. A Google Sheets node to save coding exercises and solutions.
6. A Python Code Execution node to automatically run and verify the generated Python code.

Final Workflow



Expected Result

The **AI Python Tutor** accepts Python-related questions, programming concepts, or code snippets from the user, uses the **Google Gemini Chat Model** to generate beginner-friendly explanations, clean Python code examples, debugging suggestions, and best practices, and returns an interactive learning experience. This project teaches prompt engineering, AI-assisted programming education, and conversational workflow design in n8n, making it an

excellent foundation for building intelligent coding tutors and software development learning platforms.